

# Kyla Levin

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## EDUCATION

### M.S./ Ph.D. Computer Science

University of Massachusetts Amherst, GPA: 3.97, Advisor: Yuriy Brun

Expected Graduation: May 2028

### B.S. Computer Science and Chemical Engineering

Tufts University, graduated *magna cum laude*

May 2023

## RESEARCH

### LASER Lab with Yuriy Brun

University of Massachusetts Amherst

Feb 2026 – Present

- *Py2Rocq*: Building a novel agentic formal verification system that converts existing Python code and natural language specifications into fully verified code using multiple ITP and ATP languages. Combines code generation, translation, autoformalization, and proof-writing tasks into a single pipeline.

### PLASMA Lab with Emery Berger

University of Massachusetts Amherst

Sep 2023 – Feb 2026

- *ChatDBG*: Developed a new debugging tool to converse with LLMs to reduce user involvement and make conventional debuggers more accessible to software developers. Enables LLMs to autonomously answer complex user queries about program behavior. Published and won “Distinguished Artifact” at FSE 2025: <https://dl.acm.org/doi/10.1145/3729355>
- *Pythoness*: Constructed a tool that uses LLMs to automatically generate rigorous and efficient code through natural-language descriptions and tests. Integrated an automatic testing framework that creates and runs unit tests, property-based tests, class constraints, and integration tests on generated code. Position paper: <https://arxiv.org/abs/2501.02138>

### The Foster Lab with Jeffrey Foster

Tufts University

May 2022 – May 2023

- *REST<sub>π</sub>*: Developed REST<sub>π</sub>, a novel path-sensitive type inference system that elevates REST API documentation generation by accurately capturing the relationship between API input and application and output. Helped to analyze the quality of REST API specs created with REST<sub>π</sub> implemented for Ruby built on *RDL*, an existing type-inferencing tool, against publicly used documentation software such as SwaggerHub and Postman. Published in OOPSLA 2025: <https://dl.acm.org/doi/10.1145/3763055>

### The Cowen Lab with Lenore Cowen

Tufts University

Jun – Sep 2021

- *ADAGIO*: Assisted on a graduate project to develop an efficient graph-searching algorithm that can traverse protein-protein interaction networks to identify possible unidentified disease genes. Helped evaluate on known disease modules for neurological diseases such as Alzheimer’s and Parkinson’s to locate possible clusters of causal genes. Published in ACM-BCB 2022: <https://doi.org/10.1145/3535508.3545542>

## WORK EXPERIENCE

### Applied Scientist Intern

Amazon Web Services

May – Aug 2025, May – Aug 2026

- Completed consecutive research internships at AWS with the Automated Reasoning team. Worked on developing various LLM-driven software development tools that aim to establish code correctness. Can provide references.

### Littauer Library Student Assistant Programmer

Harvard University

May – Sep 2023

- Performed full stack development on the Judaica Division’s digital collection of 8M+ records in FileMaker.
- *Front end*: Designed new web interfaces and organized a database architecture that optimized the accessibility of database navigation for people across various programming backgrounds and languages.
- *Back end*: Wrote compilation programs to better visualize collection statistics, analyze the data, and print the results into comprehensive reports.

## SKILLS

**Programming Languages:** C++, C, Java, Python, HTML5, JavaScript, Ruby

**Systems:** OpenAI, Bedrock, GDB, LLDB, PDB, Undo.io, Docker, Adobe, Git, Linux OS, LaTeX